

penev\_B — formula report

497 inline formulas (0 shown: the ones that did not render), 62 display equations, 0 tables, 17 unrecovered image regions.

Display equations

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Page | Conf.             | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| penev_B_ E00037 (5.19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 010  | ■ 0.287<br>● C +6 | $\begin{aligned} A^{\mathbf{m}}(\mathbf{x}) = & \left[ \begin{array}{c} p^l(\mathbf{x}) \quad P^M(\mathbf{x}) \end{array} \right] \left[ \begin{array}{cc} (q^{-1})^m & \mathbf{0} \\ 0 & 0 \end{array} \right] + \\ & + \bar{z} \left[ \begin{array}{c} p^l(\mathbf{x}) \quad P^M(\mathbf{x}) \end{array} \right] \left( \left[ \begin{array}{c} -u^T \\ 1 \end{array} \right] \left[ \begin{array}{cc} -u & 1 \end{array} \right] \right)^m = \\ = & \left[ \begin{array}{c} p^l(\mathbf{x}) \quad (q^{-1})^m \quad 0 \end{array} \right] + \bar{z} \left( P^M(\mathbf{x}) - p^l(\mathbf{x}) u^T \right) \left[ \begin{array}{cc} -u^m & 1 \end{array} \right] = \\ = & \left[ \begin{array}{cc} a^m(\mathbf{x}) & \mathbf{0} \end{array} \right] + \bar{z} A(\mathbf{x}) \left[ \begin{array}{cc} -u^m & 1 \end{array} \right] \end{aligned} \tag{5}$ | $A^m(\mathbf{x}) = \left[ \begin{array}{cc} p^l(\mathbf{x}) & P^M(\mathbf{x}) \end{array} \right] \left[ \begin{array}{cc} (q^{-1})^m & \mathbf{0} \\ 0 & 0 \end{array} \right] + \\ & + \bar{z} \left[ \begin{array}{cc} p^l(\mathbf{x}) & P^M(\mathbf{x}) \end{array} \right] \left( \left[ \begin{array}{c} -u^T \\ 1 \end{array} \right] \left[ \begin{array}{cc} -u & 1 \end{array} \right] \right)^m = \\ = & \left[ \begin{array}{cc} p^l(\mathbf{x}) & (q^{-1})^m \quad 0 \end{array} \right] + \bar{z} \left( P^M(\mathbf{x}) - p^l(\mathbf{x}) u^T \right) \left[ \begin{array}{cc} -u^m & 1 \end{array} \right] = \\ = & \left[ \begin{array}{cc} a^m(\mathbf{x}) & \mathbf{0} \end{array} \right] + \bar{z} A(\mathbf{x}) \left[ \begin{array}{cc} -u^m & 1 \end{array} \right]$ | $A^m(\mathbf{x}) = \left[ \begin{array}{cc} p^l(\mathbf{x}) & P^M(\mathbf{x}) \end{array} \right] \left[ \begin{array}{cc} (q^{-1})^m & \mathbf{0} \\ \mathbf{0} & 0 \end{array} \right] + \\ & + \bar{z} \left[ \begin{array}{cc} p^l(\mathbf{x}) & P^M(\mathbf{x}) \end{array} \right] \left( \left[ \begin{array}{c} -u^T \\ 1 \end{array} \right] \left[ \begin{array}{cc} -u & 1 \end{array} \right] \right)^m = \\ = & \left[ \begin{array}{cc} p^l(\mathbf{x}) & (q^{-1})^m \quad \mathbf{0} \end{array} \right] + \bar{z} \left( P^M(\mathbf{x}) - p^l(\mathbf{x}) u^T \right) \left[ \begin{array}{cc} -u^m & 1 \end{array} \right] = \\ = & \left[ \begin{array}{cc} a^m(\mathbf{x}) & \mathbf{0} \end{array} \right] + \bar{z} A(\mathbf{x}) \left[ \begin{array}{cc} -u^m & 1 \end{array} \right] \tag{5}$ |
| penev_B_ E00005 (4.62)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 001  | ■ 0.325<br>● N +0 | $\mathrm{p}_{\mathbf{l}} = \sum_{\mathbf{n}=1}^{ \mathcal{M} } \mathrm{Q}^{-1}_{\ln \mathbf{q}_{\mathbf{n}}} \cdot \mathrm{mathrm{Q}}^{\mathbf{l}} \cdot \mathrm{mathrm{q}}_{\mathbf{n}}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $p_l = \sum_{n=1}^{ \mathcal{M} } Q^{-1}_{\ln q_n}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\mathbf{p}_l = \sum_{n=1}^{ \mathcal{M} } \mathbf{Q}^{-1}_{\ln \mathbf{q}_n}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| penev_B_ E00016 (4.73)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 003  | ■ 0.397<br>● W +0 | $\begin{aligned} 0^{\mathbf{r} \text{ e } \mathbf{x}}(\mathbf{x}) = & \sum_{i=1}^{ \mathcal{M} } \mathcal{O}_i P_i(\mathbf{x}) \\ \phi^{\mathbf{r} \text{ e } \mathbf{c}}(\mathbf{x}) = & \sum_{l=1}^{ \mathcal{M} } \mathcal{O}_l K_l^{-1}(\mathbf{x}) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $O^{rex}(\mathbf{x}) = \sum_{i=1}^{ \mathcal{M} } \mathcal{O}_i P_i(\mathbf{x})$ $\phi^{rec}(\mathbf{x}) = \sum_{l=1}^{ \mathcal{M} } \mathcal{O}_l K_l^{-1}(\mathbf{x})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $O^{rec}(\mathbf{x}) = \sum_{l=1}^{ \mathcal{M} } \mathcal{O}_l P_l(\mathbf{x})$ $\phi^{rec}(\mathbf{x}) = \sum_{l=1}^{ \mathcal{M} } \mathcal{O}_l K_l^{-1}(\mathbf{x}).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| penev_B_ E00033 (5.15)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 010  | ■ 0.412<br>● W +1 | $\left[ \begin{array}{cc} -u & 1 \end{array} \right] \equiv \left[ \begin{array}{cc} -y w^{-1} & 1 \end{array} \right] = \left[ \begin{array}{cc} -p q^{-1} & 1 \end{array} \right]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\left[ \begin{array}{cc} -u & 1 \end{array} \right] \equiv \left[ \begin{array}{cc} -y w^{-1} & 1 \end{array} \right] = \left[ \begin{array}{cc} -p q^{-1} & 1 \end{array} \right]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\left[ \begin{array}{cc} -u & 1 \end{array} \right] \equiv \left[ \begin{array}{cc} -y w^{-1} & 1 \end{array} \right] = \left[ \begin{array}{cc} -p q^{-1} & 1 \end{array} \right]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| penev_B_ E00050 (5.32)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 011  | ■ 0.503<br>● W +2 | $\bar{z}^{\mathbf{l}} = P^M(\mathbf{x}_M) - a^m(\mathbf{x}_M) P^M(\mathbf{x}_m) = \mathcal{D}_{P^M}(\mathbf{x}_M).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\bar{z}^{\mathbf{l}} = P^M(\mathbf{x}_M) - a^m(\mathbf{x}_M) P^M(\mathbf{x}_m) = \mathcal{D}_{P^M}(\mathbf{x}_M).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\bar{z}^{\mathbf{l}} = P^M(\mathbf{x}_M) - a^m(\mathbf{x}_M) P^M(\mathbf{x}_m) = \mathcal{D}_{P^M}(\mathbf{x}_M).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| penev_B_ E00010 (4.67)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 001  | ■ 0.503<br>● W -1 | $0^{\mathbf{r} \text{ e } \mathbf{n}}(\mathbf{x}) = \sum_{m=1}^{ \mathcal{M} } \mathcal{O}_m a_m(\mathbf{x})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $O^{ren}(\mathbf{x}) = \sum_{m=1}^{ \mathcal{M} } \mathcal{O}_m a_m(\mathbf{x})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $O^{rec}(\mathbf{x}) = \sum_{m=1}^{ \mathcal{M} } \mathcal{O}_m a_m(\mathbf{x})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| penev_B_ E00027 (5.9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 010  | ■ 0.518<br>● N +1 | $W = \left[ \begin{array}{cc} w & x \\ y & z \end{array} \right] = \left[ \begin{array}{cc} 1_{n'} & 0 \\ y w^{-1} & 1_{n''} \end{array} \right] \left[ \begin{array}{cc} w & x \\ 0 & \bar{z}^{-1} \end{array} \right]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $W = \left[ \begin{array}{cc} w & x \\ y & z \end{array} \right] = \left[ \begin{array}{cc} 1_{n'} & 0 \\ y w^{-1} & 1_{n''} \end{array} \right] \left[ \begin{array}{cc} w & x \\ 0 & \bar{z}^{-1} \end{array} \right]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $W = \left[ \begin{array}{cc} w & x \\ y & z \end{array} \right] = \left[ \begin{array}{cc} 1_{n'} & \mathbf{0} \\ y w^{-1} & 1_{n''} \end{array} \right] \left[ \begin{array}{cc} w & x \\ \mathbf{0} & \bar{z}^{-1} \end{array} \right]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>Conf.</b> MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ &lt;0.5 — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Page | Conf.             | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Scan image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| penev_B_<br>EQ0029 (5.11)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 010  | ■ 0.537<br>●      | $\begin{aligned} \begin{aligned} W^1 &= \left[ \begin{array}{cc} w^1 & -w^1 x \tilde{z} \\ 0 & \tilde{z} \end{array} \right] \left[ \begin{array}{cc} 1 & 0 \\ -y w^{-1} & 1 \end{array} \right] = \\ &= \left[ \begin{array}{cc} w^{-1} + w^{-1} x \tilde{z} y w^{-1} & -w^{-1} x \tilde{z} \\ -\tilde{z} y w^{-1} & \tilde{z} \end{array} \right] = \\ &= \left[ \begin{array}{cc} w^{-1} & 0 \\ 0 & 0 \end{array} \right] + \left[ \begin{array}{cc} w^{-1} x \tilde{z} y w^{-1} & -w^{-1} x \tilde{z} \\ -\tilde{z} y w^{-1} & \tilde{z} \end{array} \right]. \end{aligned} \end{aligned}$ | $W^1 = \left[ \begin{array}{cc} w^1 & -w^1 x \tilde{z} \\ 0 & \tilde{z} \end{array} \right] \left[ \begin{array}{cc} 1 & 0 \\ -y w^{-1} & 1 \end{array} \right] =$ $= \left[ \begin{array}{cc} w^{-1} + w^{-1} x \tilde{z} y w^{-1} & -w^{-1} x \tilde{z} \\ -\tilde{z} y w^{-1} & \tilde{z} \end{array} \right] =$ $= \left[ \begin{array}{cc} w^{-1} & 0 \\ 0 & 0 \end{array} \right] + \left[ \begin{array}{cc} w^{-1} x \tilde{z} y w^{-1} & -w^{-1} x \tilde{z} \\ -\tilde{z} y w^{-1} & \tilde{z} \end{array} \right].$ | $W^1 = \left[ \begin{array}{cc} w^1 & -w^1 x \tilde{z} \\ 0 & \tilde{z} \end{array} \right] \left[ \begin{array}{cc} 1 & 0 \\ -y w^{-1} & 1 \end{array} \right] =$ $= \left[ \begin{array}{cc} w^{-1} + w^{-1} x \tilde{z} y w^{-1} & -w^{-1} x \tilde{z} \\ -\tilde{z} y w^{-1} & \tilde{z} \end{array} \right] =$ $= \left[ \begin{array}{cc} w^{-1} & 0 \\ 0 & 0 \end{array} \right] + \left[ \begin{array}{cc} w^{-1} x \tilde{z} y w^{-1} & -w^{-1} x \tilde{z} \\ -\tilde{z} y w^{-1} & \tilde{z} \end{array} \right].$ |
| penev_B_EQ0024 (5.6)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 009  | ■ 0.542<br>● C +4 | $\mathbf{K}_{\mathbf{j}}^{(n)} = \sum_{\mathbf{f} \in \Delta_{\mathbf{j}}} \Psi_{\mathbf{f}} \left( \frac{1}{\sigma_{\mathbf{f}}} \right)^n (\Psi_{\mathbf{f}},).$                                                                                                                                                                                                                                                                                                                                                                                                                             | $\mathbf{K}_j^{(n)} = \sum_{\mathbf{f} \in \Delta_j} \Psi_{\mathbf{f}} \left( \frac{1}{\sigma_{\mathbf{f}}} \right)^n (\Psi_{\mathbf{f}},).$                                                                                                                                                                                                                                                                                                                                                                                  | $\mathbf{K}_j^{(n)} = \sum_{\mathbf{f} \in \Delta_j} \Psi_{\mathbf{f}} \left( \frac{1}{\sigma_{\mathbf{f}}} \right)^n (\Psi_{\mathbf{f}},).$                                                                                                                                                                                                                                                                                                                                                                                  |
| penev_B_<br>EQ0013 (4.70)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 001  | ■ 0.665<br>● W +1 | $\phi^{\mathrm{rec}}(\mathbf{x}) = \sum_{m=1}^{ \mathcal{M} } O_m \sum_{n=1}^{ \mathcal{M} } \mathbf{Q}_{mn}^{-1} K^{(-1)}_n(\mathbf{x})$                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\phi^{\mathrm{rec}}(\mathbf{x}) = \sum_{m=1}^{ \mathcal{M} } O_m \sum_{n=1}^{ \mathcal{M} } \mathbf{Q}_{mn}^{-1} K^{(-1)}_n(\mathbf{x})$                                                                                                                                                                                                                                                                                                                                                                                     | $\phi^{\mathrm{rec}}(\mathbf{x}) = \sum_{m=1}^{ \mathcal{M} } O_m \sum_{n=1}^{ \mathcal{M} } \mathbf{Q}^{-1}_{mn} K^{(-1)}_n(\mathbf{x})$                                                                                                                                                                                                                                                                                                                                                                                     |
| penev_B_EQ0026 (5.8)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 009  | ■ 0.668<br>● C +4 | $\mathbf{O}_j^* \mathbf{O}_j = \mathbf{P}_j \neq \mathbf{1}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\mathbf{O}_j^* \mathbf{O}_j = \mathbf{P}_j \neq \mathbf{1}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\mathbf{O}_j^* \mathbf{O}_j = \mathbf{P}_j \neq \mathbf{1}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| penev_B_<br>EQ0053 (5.35)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 011  | ■ 0.670<br>● W +2 | $\begin{aligned} \Delta O^{\mathrm{rec}}(\mathbf{x}) &= \Delta A^m(\mathbf{x}) O(\mathbf{x}_m) = \\ &= A^M(\mathbf{x}) \begin{bmatrix} -a^m(\mathbf{x}_M) & 1 \end{bmatrix} \begin{bmatrix} O(\mathbf{x}_m) \\ O(\mathbf{x}_M) \end{bmatrix} = \\ &= A^M(\mathbf{x}) \mathcal{D}_O(\mathbf{x}_M) \end{aligned}$                                                                                                                                                                                                                                                                                | $\begin{aligned} \Delta O^{\mathrm{rec}}(\mathbf{x}) &= \Delta A^m(\mathbf{x}) O(\mathbf{x}_m) = \\ &= A^M(\mathbf{x}) \begin{bmatrix} -a^m(\mathbf{x}_M) & 1 \end{bmatrix} \begin{bmatrix} O(\mathbf{x}_m) \\ O(\mathbf{x}_M) \end{bmatrix} = \\ &= A^M(\mathbf{x}) \mathcal{D}_O(\mathbf{x}_M). \end{aligned}$                                                                                                                                                                                                              | $\begin{aligned} \Delta O^{\mathrm{rec}}(\mathbf{x}) &= \Delta A^m(\mathbf{x}) O(\mathbf{x}_m) = \\ &= A^M(\mathbf{x}) \begin{bmatrix} -a^m(\mathbf{x}_M) & 1 \end{bmatrix} \begin{bmatrix} O(\mathbf{x}_m) \\ O(\mathbf{x}_M) \end{bmatrix} = \\ &= A^M(\mathbf{x}) \mathcal{D}_O(\mathbf{x}_M). \end{aligned}$                                                                                                                                                                                                              |
| penev_B_<br>EQ0018 (4.75)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 004  | ■ 0.672<br>● W -1 | $-\tau \frac{d}{dt} \mathcal{O}(\mathbf{x}) = \mathbf{K} \phi - \mathbf{P} G(\mathcal{O})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $-\tau \frac{d}{dt} \mathcal{O}(\mathbf{x}) = \mathbf{K} \phi - \mathbf{P} G(\mathcal{O})$                                                                                                                                                                                                                                                                                                                                                                                                                                    | $-\tau \frac{d}{dt} \mathcal{O}(\mathbf{x}) = \mathbf{K} \phi - \mathbf{P} G(\mathcal{O})$                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| penev_B_<br>EQ0034 (5.16)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 010  | ■ 0.682<br>● W +1 | $Q^{-1} = \left[ \begin{array}{cc} q^{-1} & 0 \\ 0 & 0 \end{array} \right] + \bar{z} \left[ \begin{array}{c} -u^T \\ 1 \end{array} \right] \left[ \begin{array}{cc} -u & 1 \end{array} \right].$                                                                                                                                                                                                                                                                                                                                                                                               | $\mathbf{Q}^{-1} = \left[ \begin{array}{cc} q^{-1} & 0 \\ 0 & 0 \end{array} \right] + \bar{z} \left[ \begin{array}{c} -u^T \\ 1 \end{array} \right] \left[ \begin{array}{cc} -u & 1 \end{array} \right].$                                                                                                                                                                                                                                                                                                                     | $\mathbf{Q}^{-1} = \left[ \begin{array}{cc} q^{-1} & 0 \\ 0 & 0 \end{array} \right] + \bar{z} \left[ \begin{array}{c} -u^T \\ 1 \end{array} \right] \left[ \begin{array}{cc} -u & 1 \end{array} \right].$                                                                                                                                                                                                                                                                                                                     |
| penev_B_<br>EQ0046 (5.28)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 011  | ■ 0.695<br>● N +2 | $A(\mathbf{x}) = P^M(\mathbf{x}) - \mathcal{R}_{PM}^{M-1}(\mathbf{x}).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $A(\mathbf{x}) = P^M(\mathbf{x}) - \mathcal{R}_{PM}^{M-1}(\mathbf{x}).$                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $A(\mathbf{x}) = P^M(\mathbf{x}) - \mathcal{R}_{PM}^{M-1}(\mathbf{x}).$                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| penev_B_<br>EQ0002 (4.59)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 001  | ■ 0.709<br>● C +3 | $O^{\mathrm{rec}}(\mathbf{x}) = \left( \hat{\mathbf{x}}, \mathbf{O}^* \sum_{m=1}^{ \mathcal{M} } O_m \mathbf{p}_m \right) = \sum_{m=1}^{ \mathcal{M} } O_m (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{p}_m) = \sum_{m=1}^{ \mathcal{M} } O_m a_m(\mathbf{x})$                                                                                                                                                                                                                                                                                                                                     | $O^{\mathrm{rec}}(\mathbf{x}) = \left( \hat{\mathbf{x}}, \mathbf{O}^* \sum_{m=1}^{ \mathcal{M} } O_m \mathbf{p}_m \right) = \sum_{m=1}^{ \mathcal{M} } O_m (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{p}_m) = \sum_{m=1}^{ \mathcal{M} } O_m a_m(\mathbf{x})$                                                                                                                                                                                                                                                                    | $O^{\mathrm{rec}}(\mathbf{x}) = (\hat{\mathbf{x}}, \mathbf{O}^* \sum_{m=1}^{ \mathcal{M} } O_m \mathbf{p}_m) = \sum_{m=1}^{ \mathcal{M} } O_m (\hat{\mathbf{x}}, \mathbf{O}^* \mathbf{p}_m) = \sum_{m=1}^{ \mathcal{M} } O_m a_m(\mathbf{x})$                                                                                                                                                                                                                                                                                 |
| <div><div>Conf.</div><div>MathPix confidence:</div><div><div>■ ≥0.9</div><div>■ 0.5-0.9</div><div>■ &lt;0.5</div><div>— no value</div></div></div> <div><div>LaTeX source</div><div>MathPix’s reading, unchanged;</div><div>Rendered</div><div>the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</div></div> <div><div>Residual</div><div>render vs scan (inkdrill):</div><div><div>● C component</div><div>● W weak</div><div>● S stable</div><div>● N noise</div><div>● K clean</div><div>● not measured</div></div></div> <div><div>[refined: basis]</div><div>beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</div></div> |      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Page | Conf.                                         | LaTeX source                                                                                                                                                                                                                                                        | Rendered                                                                                                                                                                                                                                        | Scan image                                                                                                                                                                                                                                      |
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| penev_B_<br>EQ0009 (4.66)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 001  | <div> <div>0.710</div> <div>W +0</div> </div> | $\mathbf{a}_{\{m\}}(\mathbf{x})=\left\langle \hat{\mathbf{a}}_{\{m\}} \mathbf{x}\right\rangle ,\mathbf{a}_{\{m\}}^{\dagger}(\mathbf{x})=\left\langle \mathbf{x} \hat{\mathbf{a}}_{\{m\}}\right\rangle ,\mathbf{a}_{\{m\}}^{\dagger}\mathbf{a}_{\{n\}}=\delta _{mn}$ | $a_m(\mathbf{x})=\langle \hat{\mathbf{x}},\mathbf{O}^*\mathbf{p}_m\rangle =\sum_{n=1}^{ \mathcal{M} }\mathbf{Q}_{mn}^{-1}\left(\hat{\mathbf{x}},\mathbf{O}^*\mathbf{q}_n\right)=\sum_{n=1}^{ \mathcal{M} }\mathbf{Q}_{mn}^{-1}P_n(\mathbf{x}).$ | $a_m(\mathbf{x})=\langle \hat{\mathbf{x}},\mathbf{O}^*\mathbf{p}_m\rangle =\sum_{n=1}^{ \mathcal{M} }\mathbf{Q}_{mn}^{-1}\left(\hat{\mathbf{x}},\mathbf{O}^*\mathbf{q}_n\right)=\sum_{n=1}^{ \mathcal{M} }\mathbf{Q}_{mn}^{-1}P_n(\mathbf{x}).$ |
| penev_B_<br>EQ0062 (5.43)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 012  | <div> <div>0.779</div> <div>S -2</div> </div> | $\Delta v^r(t)=A^M(t)\times 1$                                                                                                                                                                                                                                      | $\Delta v^{rec}(t)=A^M(t)\times 1$                                                                                                                                                                                                              | $\Delta v^{rec}(t)=A^M(t)\times 1$                                                                                                                                                                                                              |
| penev_B_<br>EQ0044 (5.26)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 010  | <div> <div>0.782</div> <div></div> </div>     | $\mathcal{R}_V^{ M }(\mathbf{x})=A^m(\mathbf{x})V(\mathbf{x}_m).$                                                                                                                                                                                                   | $\mathcal{R}_V^{ M }(\mathbf{x})=A^m(\mathbf{x})V(\mathbf{x}_m).$                                                                                                                                                                               | $\mathcal{R}_V^{ M }(\mathbf{x})=A^m(\mathbf{x})V(\mathbf{x}_m).$                                                                                                                                                                               |
| penev_B_<br>EQ0061 (5.42)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 011  | <div> <div>0.825</div> <div>N +0</div> </div> | $\Delta f^r(t)=e_n=1\times \mathcal{D}_f(\mathbf{x}_M).$                                                                                                                                                                                                            | $\Delta f^{rec}(\mathbf{x}_M)=e_n=1\times \mathcal{D}_f(\mathbf{x}_M).$                                                                                                                                                                         | $\Delta f^{rec}(\mathbf{x}_M)=e_n=1\times \mathcal{D}_f(\mathbf{x}_M).$                                                                                                                                                                         |
| penev_B_<br>EQ0059 (5.40)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 011  | <div> <div>0.868</div> <div>W +1</div> </div> | $Q_{ml}=\left\langle f_{n-l}f_{n-m}\right\rangle _{m,l=0}^{M-1}$                                                                                                                                                                                                    | $Q_{ml}=\left\langle f_{n-l}f_{n-m}\right\rangle _{m,l=0}^{M-1}$                                                                                                                                                                                | $Q_{ml}=\left\langle f_{n-l}f_{n-m}\right\rangle _{m,l=0}^{M-1}$                                                                                                                                                                                |
| penev_B_<br>EQ0007 (4.64)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 001  | <div> <div>0.887</div> <div>N +0</div> </div> | $P_{\{m\}}(\mathbf{x})\equiv P_{\mathbf{x}_m}(\mathbf{x}).$                                                                                                                                                                                                         | $P_m(\mathbf{x})\equiv P_{\mathbf{x}_m}(\mathbf{x}).$                                                                                                                                                                                           | $P_m(\mathbf{x})\equiv P_{\mathbf{x}_m}(\mathbf{x}).$                                                                                                                                                                                           |
| penev_B_<br>EQ0036 (5.18)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 010  | <div> <div>0.923</div> <div>S +1</div> </div> | $A^{\{m\}}(\mathbf{x})=P^l(\mathbf{x})(\mathbf{Q}^{-1})_l^m$                                                                                                                                                                                                        | $A^m(\mathbf{x})=P^l(\mathbf{x})(\mathbf{Q}^{-1})_l^m$                                                                                                                                                                                          | $A^m(\mathbf{x})=P^l(\mathbf{x})(\mathbf{Q}^{-1})_l^m$                                                                                                                                                                                          |
| penev_B_<br>EQ0015 (4.72)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 003  | <div> <div>0.924</div> <div>N +0</div> </div> | $\mathcal{O}_l\equiv \mathbf{Q}^{-1}_{lm}O_m.$                                                                                                                                                                                                                      | $\mathcal{O}_l\equiv \mathbf{Q}^{-1}_{lm}O_m.$                                                                                                                                                                                                  | $\mathcal{O}_l\equiv \mathbf{Q}^{-1}_{lm}O_m.$                                                                                                                                                                                                  |
| penev_B_<br>EQ0045 (5.27)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 011  | <div> <div>0.925</div> <div>N -1</div> </div> | $O^r(\mathbf{x})=\mathcal{R}_O^M(\mathbf{x})$                                                                                                                                                                                                                       | $O^{rec}(\mathbf{x})=\mathcal{R}_O^M(\mathbf{x})$                                                                                                                                                                                               | $O^{rec}(\mathbf{x})=\mathcal{R}_O^M(\mathbf{x})$                                                                                                                                                                                               |
| penev_B_EQ0022 (5.4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 006  | <div> <div>0.928</div> <div>W -1</div> </div> | $N\langle S\rangle(N,\alpha)=\left\ \Phi_N^{*+}\phi(\alpha)\right\ _S^2.$                                                                                                                                                                                           | $N\langle S\rangle(N,\alpha)=\left\ \Phi_N^{*+}\phi(\alpha)\right\ _S^2.$                                                                                                                                                                       | $N\langle S\rangle(N,\alpha)=\left\ \Phi_N^{*+}\phi(\alpha)\right\ _S^2.$                                                                                                                                                                       |
| penev_B_<br>EQ0030 (5.12)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 010  | <div> <div>0.929</div> <div>W +0</div> </div> | $Q_{ml}=\left\langle P_{\{m\}}\left(\mathbf{x}_l\right)\right\rangle _{l,m=1}^M.$                                                                                                                                                                                   | $Q_{ml}=\left\langle P_m(\mathbf{x}_l)\right\rangle _{l,m=1}^M.$                                                                                                                                                                                | $Q_{ml}=\left\langle P_m(\mathbf{x}_l)\right\rangle _{l,m=1}^M.$                                                                                                                                                                                |
| penev_B_<br>EQ0043 (5.25)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 010  | <div> <div>0.933</div> <div>S +1</div> </div> | $A(\mathbf{x})=P^M(\mathbf{x})-a^m(\mathbf{x})P^M(\mathbf{x}_m).$                                                                                                                                                                                                   | $A(\mathbf{x})=P^M(\mathbf{x})-a^m(\mathbf{x})P^M(\mathbf{x}_m).$                                                                                                                                                                               | $A(\mathbf{x})=P^M(\mathbf{x})-a^m(\mathbf{x})P^M(\mathbf{x}_m).$                                                                                                                                                                               |
| penev_B_<br>EQ0054 (5.36)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 011  | <div> <div>0.941</div> <div>N +0</div> </div> | $\Delta O^r(\mathbf{x})=\frac{\mathcal{D}_{PM}(\mathbf{x})}{\mathcal{D}_{PM}(\mathbf{x}_M)}\mathcal{D}_O(\mathbf{x}_M)$                                                                                                                                             | $\Delta O^{rec}(\mathbf{x})=\frac{\mathcal{D}_{PM}(\mathbf{x})}{\mathcal{D}_{PM}(\mathbf{x}_M)}\mathcal{D}_O(\mathbf{x}_M).$                                                                                                                    | $\Delta O^{rec}(\mathbf{x})=\frac{\mathcal{D}_{PM}(\mathbf{x})}{\mathcal{D}_{PM}(\mathbf{x}_M)}\mathcal{D}_O(\mathbf{x}_M).$                                                                                                                    |
| penev_B_<br>EQ0014 (4.71)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 001  | <div> <div>0.941</div> <div>N -1</div> </div> | $O^{\text{err}}(\mathbf{x})=O(\mathbf{x})-O^{\text{rec}}(\mathbf{x}).$                                                                                                                                                                                              | $O^{\text{err}}(\mathbf{x})=O(\mathbf{x})-O^{\text{rec}}(\mathbf{x}).$                                                                                                                                                                          | $O^{\text{err}}(\mathbf{x})=O(\mathbf{x})-O^{\text{rec}}(\mathbf{x}).$                                                                                                                                                                          |
| penev_B_<br>EQ0035 (5.17)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 010  | <div> <div>0.948</div> <div>N -2</div> </div> | $O^r(\mathbf{x})=A^m(\mathbf{x})O(\mathbf{x}_m)$                                                                                                                                                                                                                    | $O^{rec}(\mathbf{x})=A^m(\mathbf{x})O(\mathbf{x}_m)$                                                                                                                                                                                            | $O^{rec}(\mathbf{x})=A^m(\mathbf{x})O(\mathbf{x}_m)$                                                                                                                                                                                            |
| penev_B_EQ0020 (5.1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 005  | <div> <div>0.950</div> <div>N +2</div> </div> | $\alpha^t=(\theta^t,S^t,X^t,Y^t).$                                                                                                                                                                                                                                  | $\alpha^t=(\theta^t,S^t,X^t,Y^t).$                                                                                                                                                                                                              | $\alpha^t=(\theta^t,S^t,X^t,Y^t).$                                                                                                                                                                                                              |
| penev_B_<br>EQ0057 (5.38)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 011  | <div> <div>0.956</div> <div>K +0</div> </div> | $\alpha^m=(q^{-1})_l^mQ_0^l$                                                                                                                                                                                                                                        | $\alpha^m=(q^{-1})_l^mQ_0^l$                                                                                                                                                                                                                    | $\alpha^m=(q^{-1})_l^mQ_0^l$                                                                                                                                                                                                                    |
| penev_B_<br>EQ0058 (5.39)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 011  | <div> <div>0.962</div> <div>W -2</div> </div> | $q_{ml}=\left\langle Q_{\{m\}}\right\rangle _{m,l=1}^{M-1}$                                                                                                                                                                                                         | $q_{ml}=\left\langle Q_{ml}\right\rangle _{m,l=1}^{M-1}$                                                                                                                                                                                        | $q_{ml}=\left\langle Q_{ml}\right\rangle _{m,l=1}^{M-1}$                                                                                                                                                                                        |
| penev_B_<br>EQ0019 (4.76)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 004  | <div> <div>0.963</div> <div>N +1</div> </div> | $\mathbf{P}\mathbf{O}=\mathbf{K}\phi.$                                                                                                                                                                                                                              | $\mathbf{P}\mathbf{O}=\mathbf{K}\phi.$                                                                                                                                                                                                          | $\mathbf{P}\mathbf{O}=\mathbf{K}\phi.$                                                                                                                                                                                                          |
| <p><b>Conf.</b> MathPix confidence: <span style="color: green;">■</span> <math>\geq 0.9</math> <span style="color: orange;">■</span> <math>0.5-0.9</math> <span style="color: red;">■</span> <math>&lt; 0.5</math> — no value</p> <p><b>LaTeX source</b> MathPix's reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span style="color: red;">●</span> C component <span style="color: orange;">●</span> W weak <span style="color: blue;">●</span> S stable <span style="color: grey;">●</span> N noise <span style="color: green;">●</span> K clean <span style="color: grey;">●</span> not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                               |                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                 |

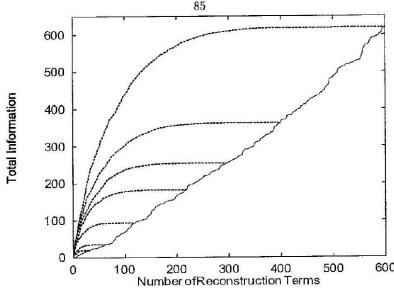
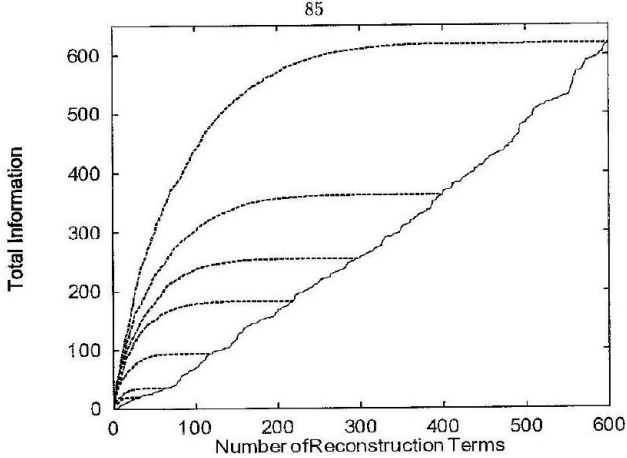
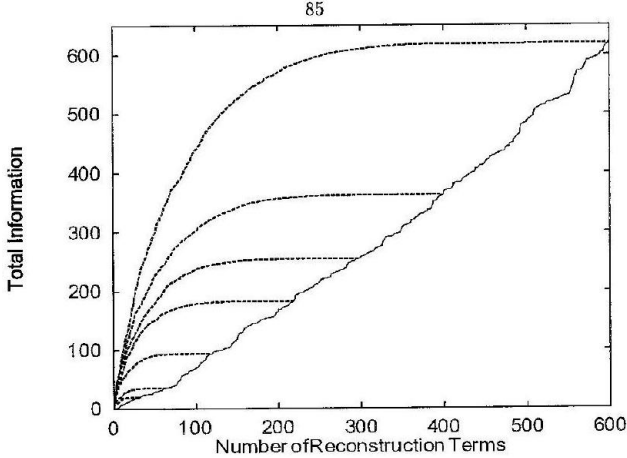
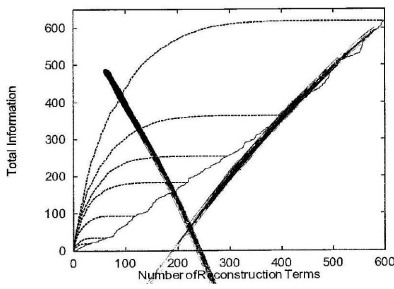
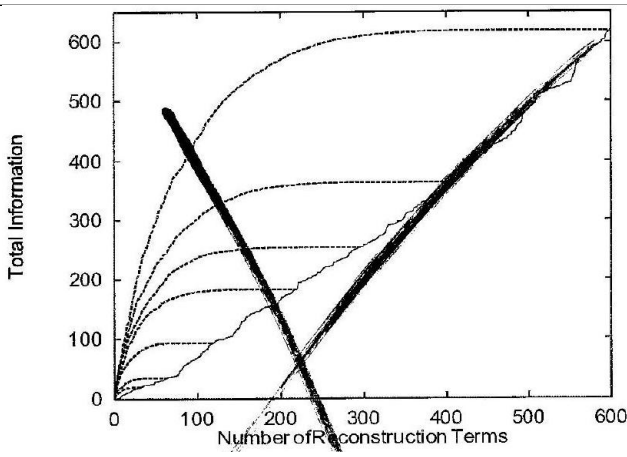
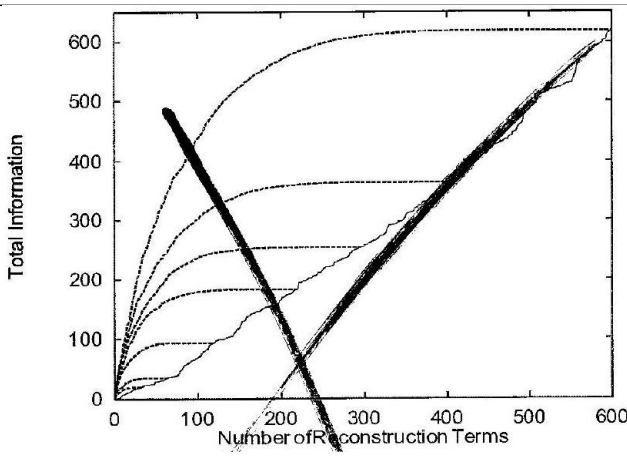
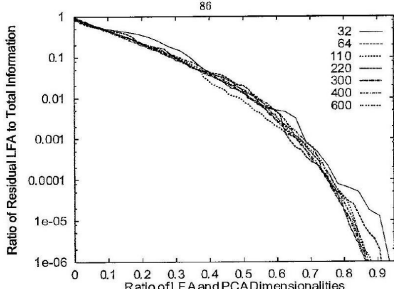
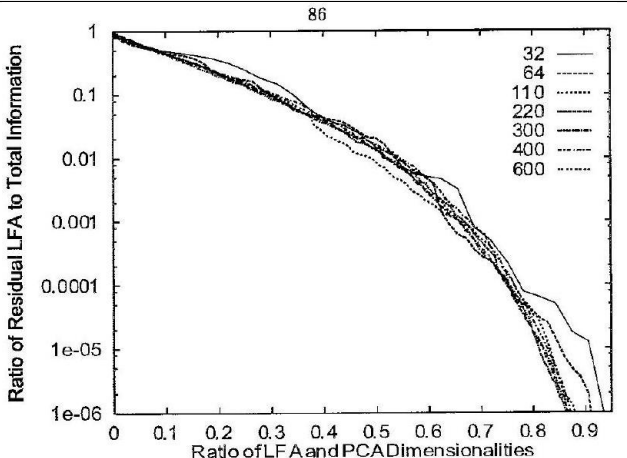
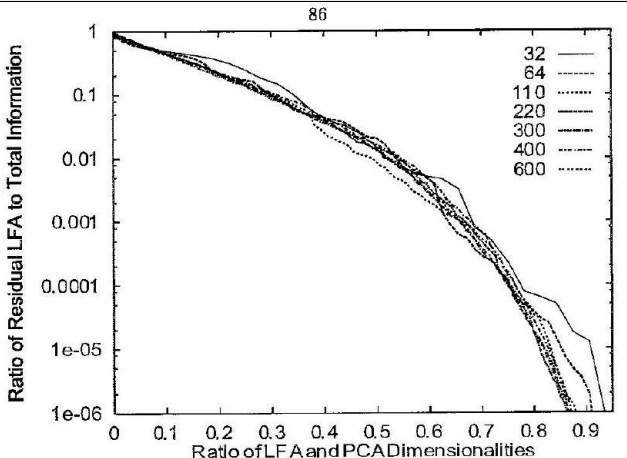
| Identifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Page | Conf.                                             | LaTeX source                                                                                                                                                                                                                                                                                                        | Rendered                                                                                                                                                                                                                                                      | Scan image                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| penev_B_<br>EQ0012 (4.69)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 001  | <div> <div>■ 0.971</div> <div>● W +2</div> </div> | $a_{\{m\}}\left(\mathrm{mathbf{x}}_{\{l\}}\right)=\sum_{n=1}^{\left \mathrm{mathcal{M}}\right }\mathrm{mathbf{Q}}^{\{-1\}}_{\{m\}n}\mathrm{P}_{\{n\}}\left(\mathrm{mathbf{x}}_{\{l\}}\right)=\sum_{n=1}^{\left \mathrm{mathcal{M}}\right }\mathrm{mathbf{Q}}^{\{-1\}}_{\{m\}n}\mathrm{mathbf{Q}}_{nl}=\delta_{ml}.$ | $a_m(\mathbf{x}_l)=\sum_{n=1}^{ \mathcal{M} }\mathbf{Q}^{-1}_{mn}P_n(\mathbf{x}_l)=\sum_{n=1}^{ \mathcal{M} }\mathbf{Q}^{-1}_{mn}\mathbf{Q}_{nl}=\delta_{ml}.$                                                                                                | $a_m(\mathbf{x}_l)=\sum_{n=1}^{ \mathcal{M} }\mathbf{Q}^{-1}_{mn}P_n(\mathbf{x}_l)=\sum_{n=1}^{ \mathcal{M} }\mathbf{Q}^{-1}_{mn}\mathbf{Q}_{nl}=\delta_{m,l}.$                                                                                               |
| penev_B_<br>EQ0028 (5.10)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 010  | <div> <div>■ 0.979</div> <div>● N +0</div> </div> | $\tilde{z}^1=z\quad yw^1x$                                                                                                                                                                                                                                                                                          | $\tilde{z}^1=z\quad yw^1x$                                                                                                                                                                                                                                    | $\tilde{z}^1=z\quad yw^1x$                                                                                                                                                                                                                                    |
| penev_B_<br>EQ0017 (4.74)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 003  | <div> <div>■ 0.981</div> <div>● W +2</div> </div> | $\sum_{l=1}^{\left \mathrm{mathcal{M}}\right }\mathrm{mathrm{Q}}_{\{m\}l}\mathrm{mathcal{O}}_{\{l\}}=0_{\{m\}}$                                                                                                                                                                                                     | $\sum_{l=1}^{ \mathcal{M} }Q_{ml}\mathcal{O}_l=O_m$                                                                                                                                                                                                           | $\sum_{l=1}^{ \mathcal{M} }\mathbf{Q}_{ml}\mathcal{O}_l=O_m$                                                                                                                                                                                                  |
| penev_B_<br>EQ0003 (4.60)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 001  | <div> <div>■ 0.981</div> <div>● N +0</div> </div> | $a_{\{m\}}(\mathrm{mathbf{x}})\equiv\left(\hat{\mathrm{mathbf{x}}},\mathrm{mathbf{O}}^*\mathrm{p}_m\right)$                                                                                                                                                                                                         | $a_m(\mathbf{x})\equiv(\hat{\mathbf{x}},\mathbf{O}^*\mathbf{p}_m)$                                                                                                                                                                                            | $a_m(\mathbf{x})\equiv(\hat{\mathbf{x}},\mathbf{O}^*\mathbf{p}_m).$                                                                                                                                                                                           |
| penev_B_<br>EQ0004 (4.61)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 001  | <div> <div>■ 0.984</div> <div>●</div> </div>      | $\delta_{m,l}=\left\langle\mathbf{q}_m,\sum_{n=1}^{\left \mathrm{mathcal{M}}\right }\mathbf{B}_{ln}\mathbf{q}_n\right\rangle=\sum_{n=1}^{\mathcal{M}}\mathbf{B}_{ln}\left\langle\mathbf{q}_m,\mathbf{q}_n\right\rangle=\sum_{n=1}^{\left \mathrm{mathcal{M}}\right }\mathbf{B}_{ln}\mathbf{Q}_{mn}$                 | $\delta_{m,l}=\left\langle\mathbf{q}_m,\sum_{n=1}^{ \mathcal{M} }\mathbf{B}_{ln}\mathbf{q}_n\right\rangle=\sum_{n=1}^{\mathcal{M}}\mathbf{B}_{ln}\left\langle\mathbf{q}_m,\mathbf{q}_n\right\rangle=\sum_{n=1}^{ \mathcal{M} }\mathbf{B}_{ln}\mathbf{Q}_{mn}$ | $\delta_{m,l}=\left\langle\mathbf{q}_m,\sum_{n=1}^{ \mathcal{M} }\mathbf{B}_{ln}\mathbf{q}_n\right\rangle=\sum_{n=1}^{\mathcal{M}}\mathbf{B}_{ln}\left\langle\mathbf{q}_m,\mathbf{q}_n\right\rangle=\sum_{n=1}^{ \mathcal{M} }\mathbf{B}_{ln}\mathbf{Q}_{mn}$ |
| penev_B_<br>EQ0011 (4.68)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 001  | <div> <div>■ 0.991</div> <div>● S +0</div> </div> | $a_{\{m\}}(\mathrm{mathbf{x}})=\sum_{n=1}^{\left \mathrm{mathcal{M}}\right }\mathrm{mathbf{Q}}^{\{-1\}}_{\{m\}n}\mathrm{P}_{\{n\}}(\mathrm{mathbf{x}}).$                                                                                                                                                            | $a_m(\mathbf{x})=\sum_{n=1}^{ \mathcal{M} }\mathbf{Q}^{-1}_{mn}P_n(\mathbf{x}).$                                                                                                                                                                              | $a_m(\mathbf{x})=\sum_{n=1}^{ \mathcal{M} }\mathbf{Q}^{-1}_{mn}P_n(\mathbf{x}).$                                                                                                                                                                              |
| penev_B_<br>EQ0041 (5.23)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 010  | <div> <div>■ 0.992</div> <div>● C +3</div> </div> | $\Delta A^{\{m\}}(\mathrm{mathbf{x}})=A^{\{M\}}(\mathrm{mathbf{x}})\left[\begin{array}{ll}-u^m&1\end{array}\right].$                                                                                                                                                                                                | $\Delta A^m(\mathbf{x})=A^M(\mathbf{x})\left[\begin{array}{ll}-u^m&1\end{array}\right].$                                                                                                                                                                      | $\Delta A^m(\mathbf{x})=A^M(\mathbf{x})\left[\begin{array}{ll}-u^m&1\end{array}\right].$                                                                                                                                                                      |
| penev_B_<br>EQ0060 (5.41)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 011  | <div> <div>■ 0.995</div> <div>● N +1</div> </div> | $e_{\{n\}}=f_{\{n\}}-\hat{f}_{\{n\}}=\mathrm{mathcal{D}}_{\{f\}}\left(\mathrm{mathbf{x}}_{\{M\}}\right)$                                                                                                                                                                                                            | $e_n=f_n-\hat{f}_n=\mathcal{D}_f(\mathbf{x}_M)$                                                                                                                                                                                                               | $e_n=f_n-\hat{f}_n=\mathcal{D}_f(\mathbf{x}_M)$                                                                                                                                                                                                               |
| penev_B_<br>EQ0047 (5.29)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 011  | <div> <div>■ 0.999</div> <div>● N +2</div> </div> | $\mathrm{mathcal{D}}_{\{V\}}(\mathrm{mathbf{x}})=V(\mathrm{mathbf{x}})-\mathrm{mathcal{R}}_{\{V\}}^{\left \mathrm{mathcal{M}}\right -1}(\mathrm{mathbf{x}})$                                                                                                                                                        | $\mathcal{D}_V(\mathbf{x})=V(\mathbf{x})-\mathcal{R}_V^{ \mathcal{M} -1}(\mathbf{x})$                                                                                                                                                                         | $\mathcal{D}_V(\mathbf{x})=V(\mathbf{x})-\mathcal{R}_V^{ \mathcal{M} -1}(\mathbf{x})$                                                                                                                                                                         |
| penev_B_<br>EQ0006 (4.63)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 001  | <div> <div>■ 0.999</div> <div>● N +0</div> </div> | $\mathrm{mathbf{Q}}_{\{n\}m}=\left\langle\mathbf{q}_n,\mathbf{q}_m\right\rangle=\left\langle\mathbf{O}\hat{\mathbf{x}}_n,\mathbf{q}_m\right\rangle=P_m(\mathbf{x}_n)$                                                                                                                                               | $\mathbf{Q}_{nm}=\langle\mathbf{q}_n,\mathbf{q}_m\rangle=\langle\mathbf{O}\hat{\mathbf{x}}_n,\mathbf{q}_m\rangle=P_m(\mathbf{x}_n)$                                                                                                                           | $\mathbf{Q}_{nm}=\langle\mathbf{q}_n,\mathbf{q}_m\rangle=\langle\mathbf{O}\hat{\mathbf{x}}_n,\mathbf{q}_m\rangle=P_m(\mathbf{x}_n)$                                                                                                                           |
| penev_B_<br>EQ0042 (5.24)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 010  | <div> <div>■ 0.999</div> <div>● S +0</div> </div> | $u^{\{m\}}=p^{\{l\}}\left(\mathrm{mathbf{x}}_{\{M\}}\right)\left(q^{-1}\right)_l^m=a^m(\mathbf{x}_M).$                                                                                                                                                                                                              | $u^m=p^l(\mathbf{x}_M)\left(q^{-1}\right)_l^m=a^m(\mathbf{x}_M).$                                                                                                                                                                                             | $u^m=p^l(\mathbf{x}_M)(q^{-1})_l^m=a^m(\mathbf{x}_M).$                                                                                                                                                                                                        |
| penev_B_<br>EQ0001 (4.58)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 001  | <div> <div>■ 1.000</div> <div>● W +1</div> </div> | $s^{\{r\ e\ c\}}=s_{\parallel}=\sum_{m=1}^{\left \mathrm{mathcal{M}}\right }O_m p_m$                                                                                                                                                                                                                                | $s^{rec}=s_{\parallel}=\sum_{m=1}^{ \mathcal{M} }O_m p_m$                                                                                                                                                                                                     | $\mathbf{s}^{rec}=\mathbf{s}_{\parallel}=\sum_{m=1}^{ \mathcal{M} }O_m\mathbf{p}_m.$                                                                                                                                                                          |
| penev_B_<br>EQ0031 (5.13)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 010  | <div> <div>■ 1.000</div> <div>● W +0</div> </div> | $p^{\{T\}}_{\{l\}}\equiv y^{\{T\}}_{\{l\}}=x_{\{l\}}=Q_{Ml}=P_M(\mathbf{x}_l) _{l=1}^{M-1}.$                                                                                                                                                                                                                        | $p^T_l\equiv y^T_l=x_l=Q_{Ml}=P_M(\mathbf{x}_l) _{l=1}^{M-1}.$                                                                                                                                                                                                | $p^T_l\equiv y^T_l=x_l=Q_{Ml}=P_M(\mathbf{x}_l) _{l=1}^{M-1}.$                                                                                                                                                                                                |
| penev_B_<br>EQ0008 (4.65)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 001  | <div> <div>■ 1.000</div> <div>● K +0</div> </div> | $\mathrm{mathbf{Q}}=\mathrm{P} _{\mathrm{mathcal{M}}}.$                                                                                                                                                                                                                                                             | $\mathbf{Q}=\mathbf{P} _{\mathcal{M}}.$                                                                                                                                                                                                                       | $\mathbf{Q}=\mathbf{P} _{\mathcal{M}}.$                                                                                                                                                                                                                       |
| penev_B_EQ0021 (5.2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 005  | <div> <div>■ 1.000</div> <div>● N +0</div> </div> | $\phi^{\{t\}}=H\left(I^{\{t\}}\;;\;\alpha^{\{t\}}\right)$                                                                                                                                                                                                                                                           | $\phi^t=H\left(I^t;\alpha^t\right)$                                                                                                                                                                                                                           | $\phi^t=H(I^t;\alpha^t)$                                                                                                                                                                                                                                      |
| penev_B_EQ0023 (5.5)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 007  | <div> <div>■ 1.000</div> <div>● N +0</div> </div> | $\alpha_h^{\{t\}}=\left(\theta^t,(1+h)S^t,X^t,Y^t\right)$                                                                                                                                                                                                                                                           | $\alpha_h^t=(\theta^t,(1+h)S^t,X^t,Y^t)$                                                                                                                                                                                                                      | $\alpha_h^t=(\theta^t,(1+h)S^t,X^t,Y^t)$                                                                                                                                                                                                                      |
| penev_B_EQ0025 (5.7)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 009  | <div> <div>■ 1.000</div> <div>● N -1</div> </div> | $O_{\{j\}}=\mathrm{mathbf{K}}_{\{j\}}\phi_{\{j\}}=\mathrm{mathbf{K}}_j\mathrm{P}_{\{j\}}\phi=\mathrm{mathbf{K}}_j\phi$                                                                                                                                                                                              | $O_j=\mathbf{K}_j\phi_j=\mathbf{K}_j\mathbf{P}_j\phi=\mathbf{K}_j\phi$                                                                                                                                                                                        | $O_j=\mathbf{K}_j\phi_j=\mathbf{K}_j\mathbf{P}_j\phi=\mathbf{K}_j\phi$                                                                                                                                                                                        |
| penev_B_<br>EQ0032 (5.14)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 010  | <div> <div>■ 1.000</div> <div>● N +0</div> </div> | $z=\mathrm{P}_{\{M\}}\left(\mathrm{mathbf{x}}_{\{M\}}\right).$                                                                                                                                                                                                                                                      | $z=P_M(\mathbf{x}_M).$                                                                                                                                                                                                                                        | $z=P_M(\mathbf{x}_M).$                                                                                                                                                                                                                                        |
| penev_B_<br>EQ0038 (5.20)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 010  | <div> <div>■ 1.000</div> <div>● N +2</div> </div> | $A(\mathrm{mathbf{x}})\equiv P^{\{M\}}(\mathrm{mathbf{x}})-p^{\{l\}}(\mathrm{mathbf{x}})u^{\{T\}}_{\{l\}}$                                                                                                                                                                                                          | $A(\mathbf{x})\equiv P^M(\mathbf{x})-p^l(\mathbf{x})u^T_l$                                                                                                                                                                                                    | $A(\mathbf{x})\equiv P^M(\mathbf{x})-p^l(\mathbf{x})u^T_l.$                                                                                                                                                                                                   |
| <p><b>Conf.</b> MathPix confidence: <span>■ ≥0.9</span> <span>■ 0.5–0.9</span> <span>■ &lt;0.5</span> — no value</p> <p><b>LaTeX source</b> MathPix’s reading, unchanged; <b>Rendered</b> the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ</p> <p><b>Residual</b> render vs scan (inkdrill): <span>● C</span> component <span>● W</span> weak <span>● S</span> stable <span>● N</span> noise <span>● K</span> clean <span>●</span> not measured</p> <p><b>[refined: <i>basis</i>]</b> beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.</p> |      |                                                   |                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                               |

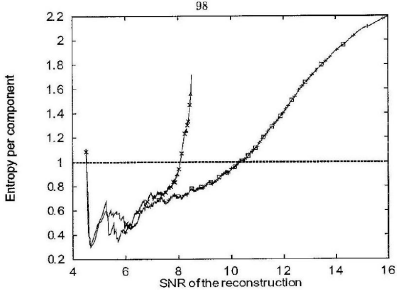
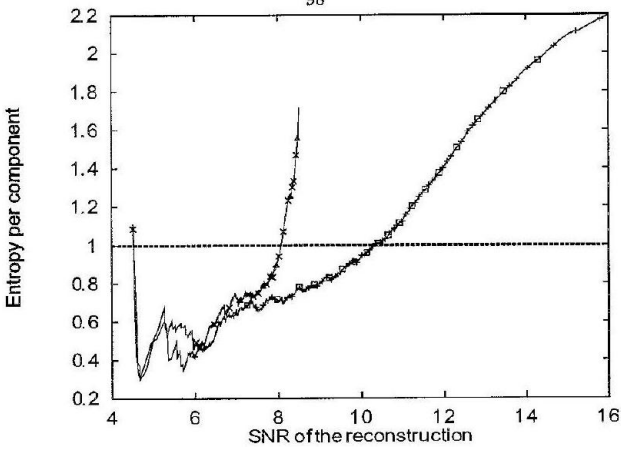
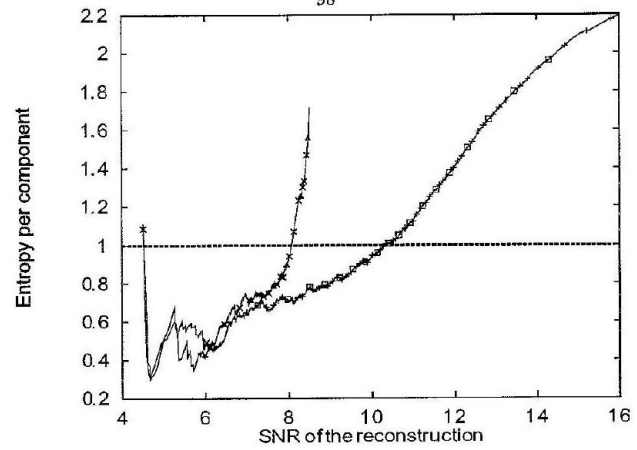
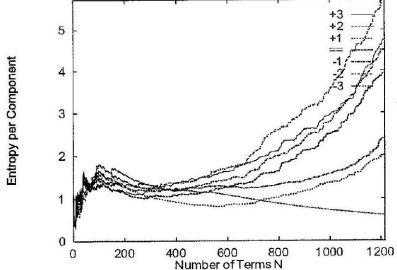
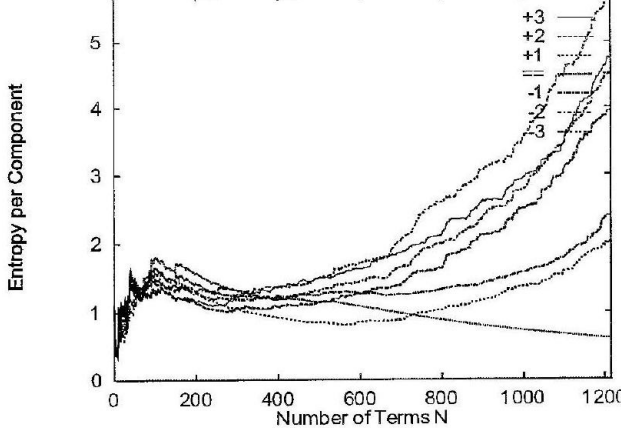
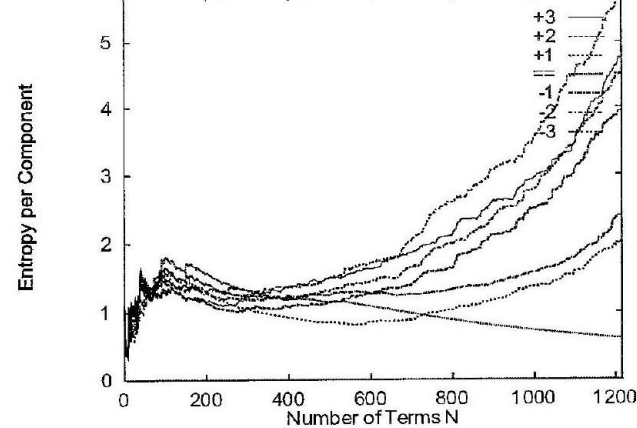
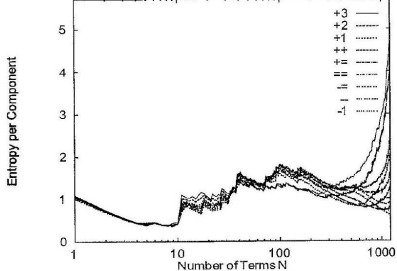
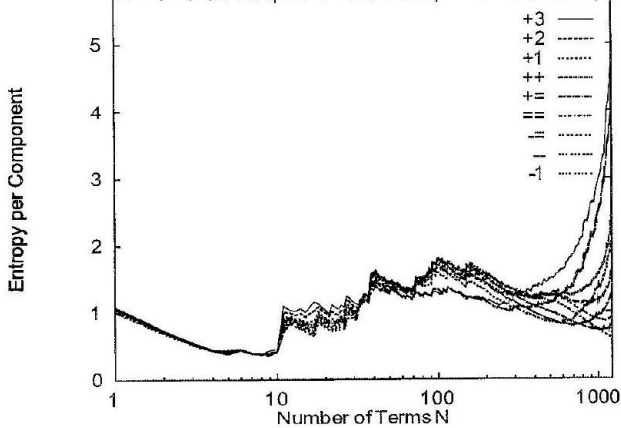
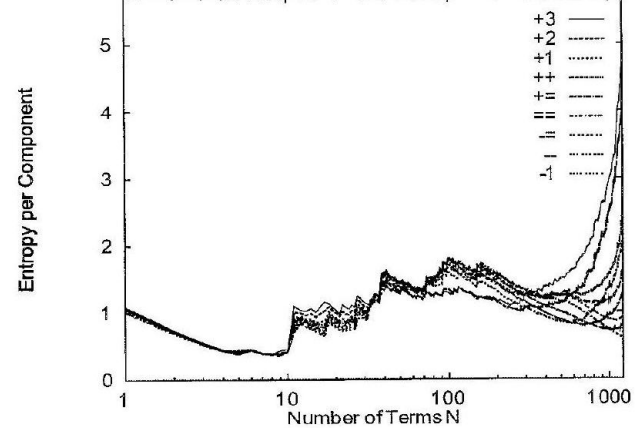
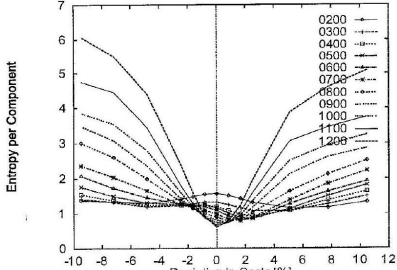
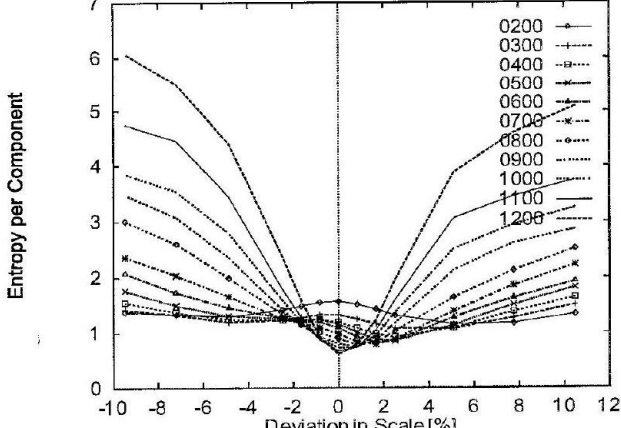
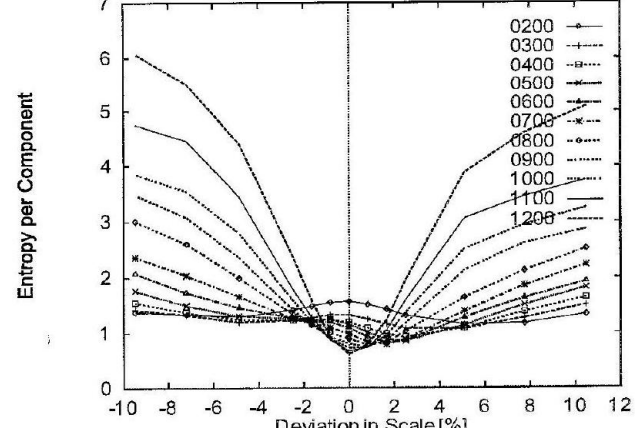


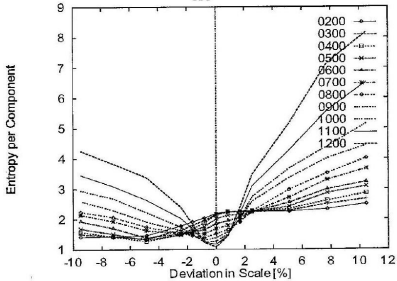
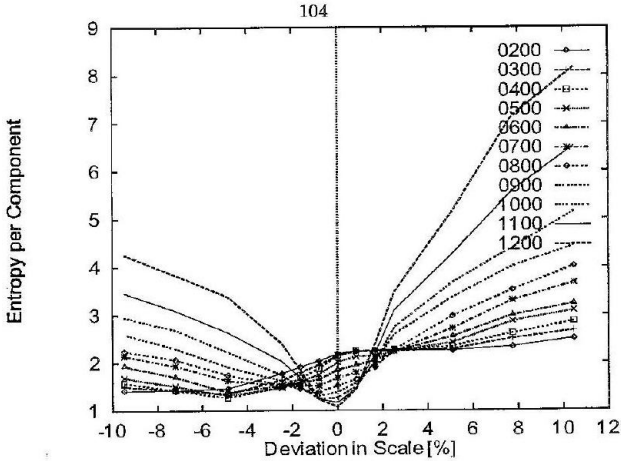
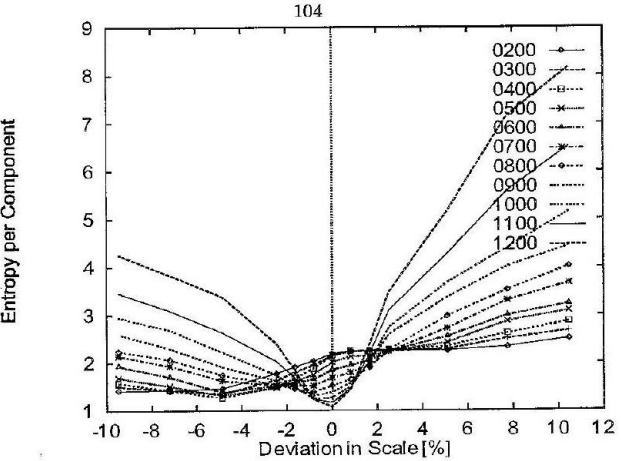
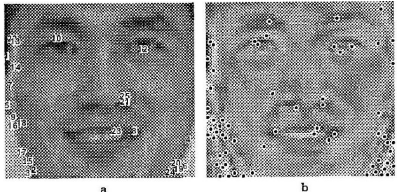
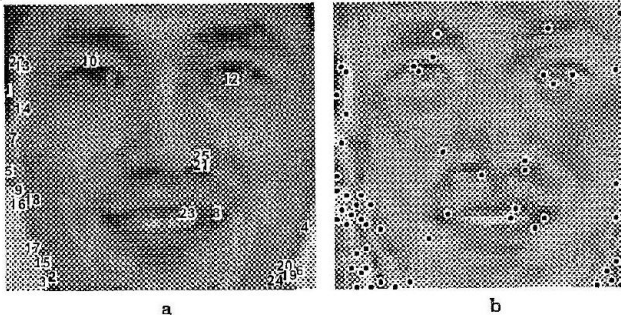
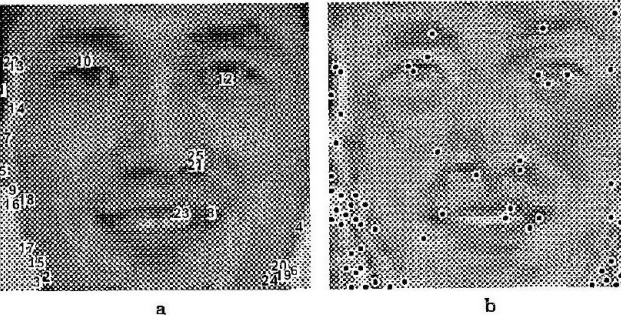
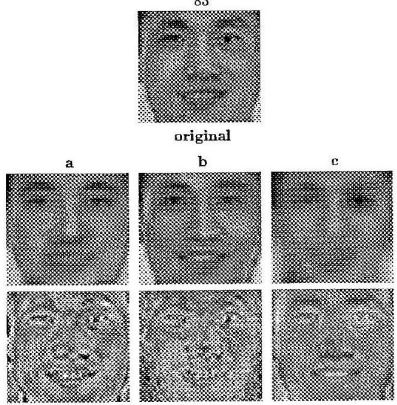
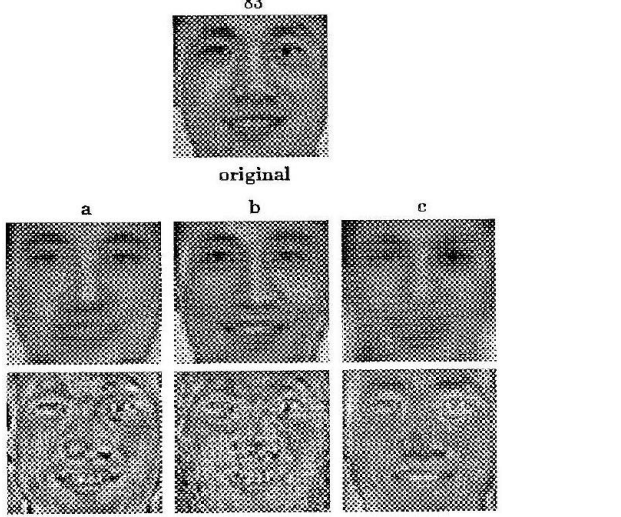
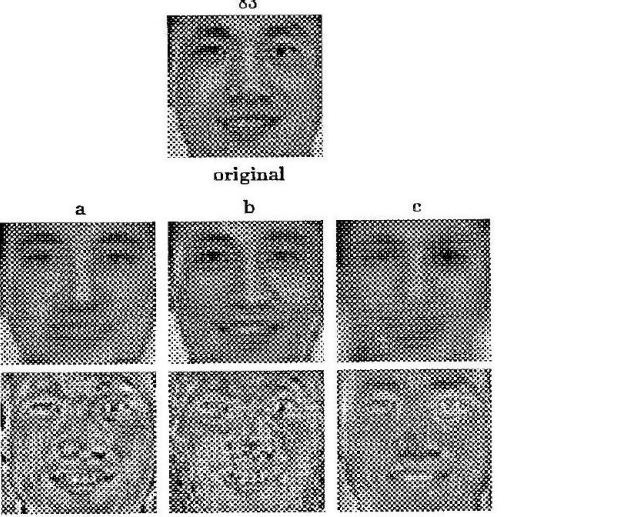


Image regions — rendered against scan

Two comparable cells per row. **Rendered**: the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

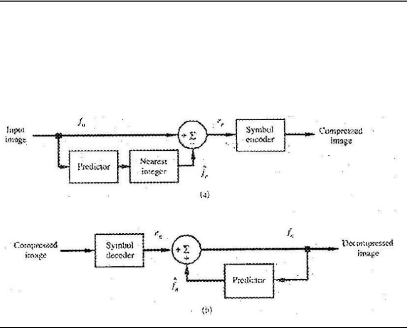
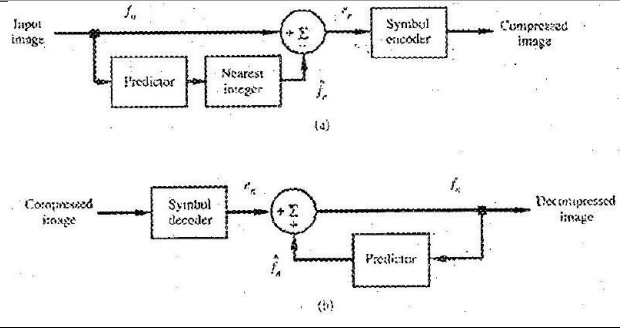
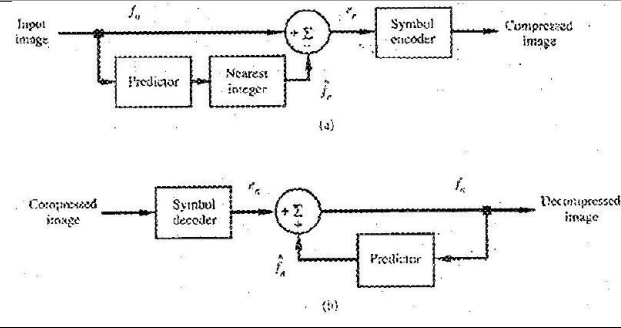
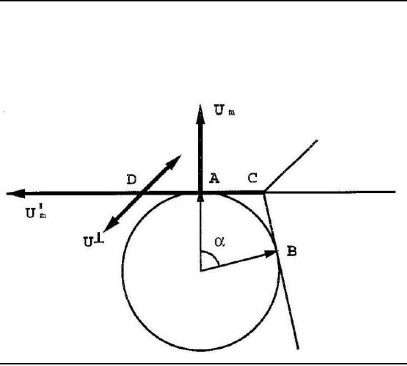
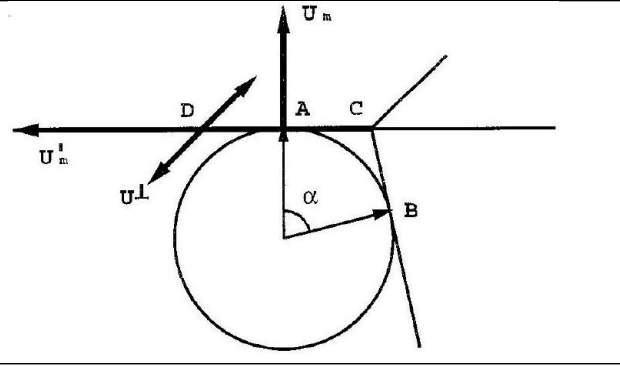
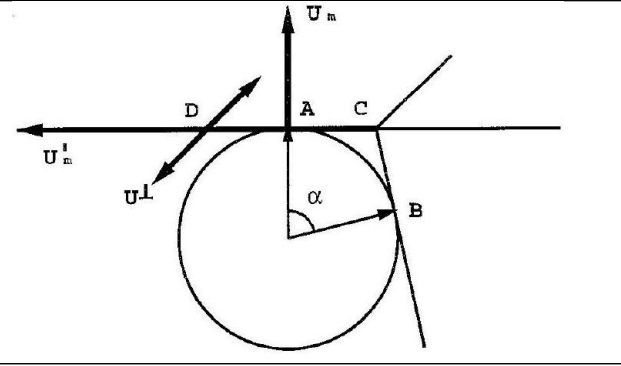
| Identifier       | Page | Class   | Source                                                        | Author source                                                                         | Rendered                                                                              | Scan                                                                                  |
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|                  |     |         |                                                              |                                                                                       |                                                                                       |                                                                                       |
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| penev_B_PIC_0007 | 007 | — (dup) | 5fff204a-b675-4b41-925e-7cb5ba4e9fe4-07_580_809_91_1065.jpg  |  |  |  |

|                  |     |         |                                                               |                                                                                       |                                                                                       |                                                                                       |
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| penev_B_DIA_0001 | 002 | — (dup) | 5fff204a-b675-4b41-925e-7cb5ba4e9fe4-02_413_841_110_110.jpg   |    |    |    |
| penev_B_DIA_0002 | 002 | — (dup) | 5fff204a-b675-4b41-925e-7cb5ba4e9fe4-02_685_651_1435_217.jpg  |  |   |   |
| penev_B_DIA_0003 | 003 | — (dup) | 5fff204a-b675-4b41-925e-7cb5ba4e9fe4-03_418_813_94_1046.jpg   |  |  |  |



|                  |     |         |                                                              |  |  |  |
|------------------|-----|---------|--------------------------------------------------------------|--|--|--|
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|                  |     |         |                                                              |                                                                                     |                                                                                     |                                                                                     |
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